Lesson plan · for the teacher

The best for whom?

Optimization · Level 1 · Lesson 4

Time
40 min

Materials
Board and chalk; A list of five class duties; Small paper slips for pupils to score each duty from 1 to 5

Learning goals

- Judge a plan by its total, its worst-off and its gap.
- Compare first come, first served with other sharing rules.
- Agree a fair rule for class duties together.

Activity: Fair class duties

1. Warm-up (5 min): Ask which class job nobody wants and why. Write five duties on the board, including that one.
2. Score (7 min): In groups of five, each pupil writes a score from 1 (hate it) to 5 (happy to do it) for each duty on a slip of paper.
3. Two plans (13 min): Each group makes two plans. First, pupils choose in turn by alphabetical order. Second, they try to make the lowest score in the group as high as possible.
4. Compare (8 min): Groups write each plan's total and lowest score on the board. Discuss: which plan is better overall, and which is kinder to the worst-off?
5. Agree (7 min): The class votes on a rule for next month and agrees how the hardest job will rotate. Write the rule on the wall.

Keep the scores private within each group so nobody is teased for disliking a job. The point is the rule, not who said what.

Discussion

- Who came off worst under first come, first served?
- Is a rule fair just because everyone knows it?
- How should we share the job nobody wants?

Exit ticket

- Write one sharing rule at our school, who it is hard on, and one way to make it fairer.

Try it in class tomorrow
Plan class duties fairly: pupils score each job from 1 to 5. Compare first come, first served with a plan that lifts the lowest score, and rotate the hardest job. Ask the class which plan they would choose, and why.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Amina plans class duties. The plan with the best total score gives Musa the job nobody wants (cleaning the latrines) every single week. Is it fair?**

- ◯ Yes: the total is the best, so it's fair
- ◯ Not necessarily: check the worst-off, and share the hard job
- ◯ Only if that pupil agreed once

2 **Which of these help you judge whether a sharing plan is fair? Choose all that apply.**

- ☐ Who happened to choose first
- ☐ How quickly the plan was made
- ☐ The lowest score anyone gets
- ☐ The total of everyone's scores
- ☐ The gap between the best and the worst

3 **In the teachers-to-classes example, the plan with the best total (39) gives every teacher a score at least as high as first come, first served did.**

- ◯ True
- ◯ False

4 **Put in order: agreeing a fair way to share class duties.**

- ☐ Collect everyone's scores, and let pupils correct mistakes
- ☐ Review it after a few weeks, and rotate the hardest job
- ☐ Show the plan and its numbers to everyone
- ☐ Agree the rule with the class before anyone sees the result
- ☐ Make the plan with the agreed rule

5 **Two plans for exam rooms. Plan A: total 30, lowest score 2. Plan B: total 28, lowest score 5. Which plan looks after the worst-off better?**

- ◯ Plan B: its lowest score is higher
- ◯ They are the same
- ◯ Plan A: its total is higher

Answers · for the teacher

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1 **Not necessarily: check the worst-off, and share the hard job**

The total can hide one person who always loses.

2 **The total of everyone's scores; The lowest score anyone gets; The gap between the best and the worst**

The total, the worst-off and the gap together show how good a plan is and who carries its cost.

3 **False**

Most teachers gain, and Karim gains most (2 → 9), but Amina drops from 9 (maths) to 6 (history).

4 **Top to bottom: 2, 5, 4, 1, 3**

Agreeing the rule first stops people choosing the rule that suits them once they see the result.

5 **Plan B: its lowest score is higher**

Plan B lifts the lowest score from 2 to 5, at a cost of 2 points of total.

For you to think about

Think of something your school shares out: classes, duties, rooms, the good periods. Who comes off worst under the rule you use now, and would they agree it's fair?